

Harsh Patil

Pune, Maharashtra

Data Science | Machine Learning | Generative AI | Data Analytics



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SUMMARY

Recently graduated with a Master's degree in Data Science, equipped with a strong foundation in Python, SQL, Machine Learning, Generative AI, and Data Analytics. Hands on experience building AI-driven and data analytics projects through academic work. Eager to apply my technical skills, analytical mindset, and problem-solving abilities in an industrial environment while contributing to innovative, data-driven solutions.

TECHNICAL & SOFT SKILLS

- **Programming & Tools:** Python, SQL, MySQL, Power BI, GitHub, Claude Code.
- **Data & Machine Learning:** Pandas, NumPy, Scikit-learn, Data Cleaning & Preprocessing, Predictive Analytics.
- **GenAI :** LLMs, Generative AI, RAG, FAISS, BM25, Prompt Engineering, Groq API.
- **Development & Deployment:** FastAPI, Streamlit, GitHub Actions.
- **Soft Skills :** Technical Documentation, Problem Solving, Collaboration Skills.

PROJECTS

Ask My Docs – Domain Specific RAG System with Automated Citation Verification | [GitHub](#)

Python | RAG | LLMs | FAISS | BM25 | FastAPI | Streamlit | Groq API | Claude Code

- Engineered a hybrid RAG pipeline combining FAISS vector search and BM25, fused via Reciprocal Rank Fusion (RRF) and cross-encoder reranking over structure-aware PDF chunks preserving sections and tables.
- Built automated citation verification to validate source references, flag uncited claims, detect unsupported statements, and refuse responses when required information is absent from the documents.
- Shipped a FastAPI + Streamlit application with 561 offline tests and 89% code coverage, integrated with GitHub Actions to detect regressions in retrieval and citation accuracy against established baselines.

Vendor Invoice Intelligence Platform – ML System for Freight Cost Prediction & Invoice Risk Flagging | [GitHub](#)

Python | Scikit-learn | Pandas | SQLite | Streamlit | Plotly

- Built an end-to-end ML pipeline predicting vendor freight costs (regression) and flagging high risk invoices (classification), achieving 96.99% R^2 and 89% accuracy / 0.96 precision using GridSearchCV tuned Random Forest models.
- Deployed an interactive Streamlit + Plotly dashboard with live risk probability gauges, giving finance teams real-time cost estimates and auto-approval decisions with no code required.
- Engineered a modular training/inference architecture with joblib model persistence, enabling reproducible retraining and easy model swaps across both prediction modules.

AI-Powered Synthetic Data Generator & Data Cleaning Tutor | [GitHub](#)

Python | Streamlit | Pandas | Faker | Groq API | LLM | GenAI

- Built a data quality simulation engine generating datasets with varying levels of missing values, duplicates, formatting errors, and standardization issues to practice data cleaning workflows.
- Developed an automated module that generates executable Python and SQL scripts to handle missing data, duplicates, and formatting inconsistencies, improving data accuracy and consistency.
- Documented data-quality rules and cleaning logic to support repeatable, well-organized data processes.

EDUCATION

MIT World Peace University | Pune

August 2024 - May 2026

Master's in Data Science & Big Data Analytics - CGPA 8.49

Modern College, Ganeshkhind | Pune

September 2020 - April 2023

Bachelor of Science in Computer Science - CGPA 8.01

CERTIFICATIONS

- **IBM - Python for Data Science, AI & Development.**
- **ORACLE - Agentic AI Certified Foundations Associate.**
- **IBM - Generative AI: Prompt Engineering.**
- **Tata - GenAI Powered Data Analytics Job Simulation.**